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INTERVERTEBRAL SPINAL IMPLANT

Abstract

An intervertebral implant for implantation in an intervertebral space between vertebrae. The implant includes a body extending from an upper surface to a lower surface. The body has a front end, a rear end and a pair of spaced apart first and second side walls extending between the front and rear walls such that an interior chamber is defined within the front and rear ends and the first and second walls. The body defines an outer perimeter and an inner perimeter extending about the internal chamber. At least one of the side walls is defined by a solid support structure and an integral porous structure, the porous structure extending from the outer perimeter to the inner perimeter. The porous structure embeds or encapsulates at least a portion of the solid support structure.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/933,560, filed on Sep. 20, 2022, which is a continuation of U.S. patent application Ser. No. 16/600,632 filed on Oct. 14, 2019 (published as U.S. Pat. Pub. No. 2020-0038198), which is a continuation of U.S. patent application Ser. No. 15/973,756 filed on May 8, 2018, now U.S. Pat. No. 10,555,819, which is a continuation of U.S. patent application Ser. No. 15/973,609 filed May 8, 2018, now U.S. Pat. No. 11,141,286, all of which are incorporated by reference herein in their entireties for all purposes.

FIELD

[0002] The present disclosure generally relates to fixation devices and systems for positioning and immobilizing at least two adjacent vertebrae and methods related to the same. In particular, the present disclosure relates to interbody fusion devices with an integrated solid support structure and porous ingrowth structure.

BACKGROUND

[0003] The spine is the axis of the skeleton on which all of the body parts “hang”. In humans, the normal spine has seven cervical, twelve thoracic and five lumbar segments. The lumbar spine sits upon the sacrum, which then attaches to the pelvis, and in turn is supported by the hip and leg bones. The bony vertebral bodies of the spine are separated by intervertebral discs, which act as joints but allow known degrees of flexion, extension, lateral bending, and axial rotation.

[0004] The typical vertebra has a thick anterior bone mass called the vertebral body, with a neural (vertebral) arch that arises from the posterior surface of the vertebral body. The central of adjacent vertebrae are supported by intervertebral discs. The spinal disc and/or vertebral bodies may be displaced or damaged due to trauma, disease, degenerative defects, or wear over an extended period of time. One result of this displacement or damage to a spinal disc or vertebral body may be chronic back pain. In many cases, to alleviate back pain from degenerated or herniated discs, the disc is removed along with all or part of at least one neighboring vertebrae and is replaced by an implant that promotes fusion of the remaining bony anatomy.

[0005] However, the success or failure of spinal fusion may depend upon several factors. For instance, the spacer or implant or cage used to fill the space left by the removed disc and bony anatomy must be sufficiently strong to support the spine under a wide range of loading conditions. The spacer should also be configured so that it likely to remain in place once it has been positioned in the spine by the surgeon. Additionally, the material used for the spacer should be biocompatible material and should have a configuration that promotes bony ingrowth.

SUMMARY

[0006] To meet this and other needs, intervertebral implants for use with the anterior, antero-lateral, lateral, and/or posterior portions of at least one motion segment unit of the spine, systems, and

methods are provided. Traditionally, interbody spacers or implants intended to help facilitate intervertebral fusion may serve as a means to restore intervertebral height and/or lordosis. The implant may feature a central lumen to house bone graft material. It is through this central lumen where most of the fusion may occur. The implants of the disclosure incorporate a volumetric, interconnected porosity throughout the entire spacer. This enables bone to grow into and/or through the spacer, making it part of the fusion mass. The incorporation of a volumetric, interconnected porosity within the implant may encourage faster, stronger intervertebral fusion.

[0007] According to one embodiment, a cervical intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body having a front end, a rear end and opposed side walls extending between the ends. The body has an outer perimeter with a trapezoidal shape and an inner perimeter about an internal chamber. The body includes an upper surface and a lower surface with the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim. A solid front wall extends at the front end between at least the solid upper outer rim and the solid lower outer rim. A solid rear wall extends at the rear end between at least the solid upper outer rim and the solid lower outer rim. Each of the side walls includes at least one solid cross strut extending between the solid upper rims and at least one solid cross strut extending between the solid lower rims. Each of the side walls includes at least one solid support structure extending between the upper surface and the lower surface with the solid support structure occupying a minimal space within each side wall. A porous structure is integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and each of the solid structures in each of the side walls and extends from the body outer perimeter to the body inner perimeter.

[0008] According to another embodiment, an intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body having a front end, a rear end and opposed side walls extending between the ends. The body has an outer perimeter and an inner perimeter about an internal chamber. The body includes an upper surface and a lower surface. The upper surface is defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface is defined by a solid lower outer rim and a spaced apart solid lower inner rim. A solid front wall extends at the front end between at least the solid upper outer rim and the solid lower outer rim. A solid rear wall extends at the rear end between at least the solid upper outer rim and the solid lower outer rim. Each of the side walls includes at least one solid support structure extending between the upper and lower surfaces. A porous structure is integrally formed with the solid upper rims, the solid lower rims and the at least one solid support structure in each of the side walls. The porous structure extends from the body outer perimeter to the body inner perimeter while the solid upper and lower outer rims and the solid front and rear walls extend along the outer perimeter such that the porous structure is encased within solid structure.

[0009] According to yet another embodiment, a cervical intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body extending from an upper surface to a lower surface. The body has a front end, a rear end and a pair of spaced apart first and second side walls extending between the front and rear walls such that an interior chamber is defined within the front and rear ends and the first and second walls. The body defines an outer perimeter with a trapezoidal shape and an inner perimeter extending about the internal chamber. At least one of the walls is defined by a solid support structure and an integral porous structure. The porous structure extends from the outer perimeter to the inner perimeter and from the upper surface to the lower surface and encapsulates at least a portion of the solid support structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A more complete understanding of the present disclosure, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

[0011] FIGS. **1-4** are perspective, side, top and rear views, respectively, of an intervertebral implant according to one embodiment of the disclosure with the porous portions shown textured;

[0012] FIGS. **5-7** are perspective, top and side views, respectively, of the intervertebral implant of FIGS. **1-4** with the porous portions removed to show the support structure;

[0013] FIGS. **8-10** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0014] FIGS. **11-13** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0015] FIGS. **14-16** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0016] FIGS. **17-19** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0017] FIGS. **20-22** and **24** are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured, and FIG. **23** is a cross-sectional view along the lines **23-23** in FIG. **21**;

[0018] FIGS. **25-27** are perspective, top and side views, respectively, of the intervertebral implant of FIGS. **20-24** with the porous portions removed to show the support structure;

[0019] FIGS. **28-31** are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0020] FIGS. **32-34** are perspective, top and side views, respectively, of the intervertebral implant of FIGS. **28-31** with the porous portions removed to show the support structure;

[0021] FIGS. **35** and **36** are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0022] FIGS. **37** and **38** are rear and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0023] FIG. **39** is a perspective view of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0024] FIGS. **40-43** are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0025] FIGS. **44-47** are perspective, top, side and rear views, respectively, of the intervertebral implant of FIGS. **40-43** with the porous portions removed to show the support structure;

[0026] FIGS. **48-51** are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0027] FIGS. **52** and **53** are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0028] FIGS. **54-57** are perspective, rear, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0029] FIGS. **58-61** are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0030] FIGS. **62-64** are perspective, top and side views, respectively, of the intervertebral implant

of FIGS. **58-61** with the porous portions removed to show the support structure;

[0031] FIGS. **65-67** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0032] FIGS. **68** and **69** are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0033] FIGS. **70-72** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0034] FIGS. **73-75** are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0035] FIGS. **76** and **77** are perspective and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0036] FIGS. **78** and **79** are perspective and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0037] FIGS. **80-83** are front perspective, rear perspective, top and side views, respectively, of the spacer portion of the implants of FIGS. **76-79** with the porous portions shown textured;

[0038] FIGS. **84-86** are perspective, top and side views, respectively, of the spacer portion of FIGS. **80-83** with the porous portions removed to show the support structure; and

[0039] FIGS. **87** and **88** are illustrative photos of various porous structures in accordance with embodiments of the disclosure;

[0040] FIGS. **89-92** are perspective, bottom, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0041] FIGS. **93-96** are perspective, bottom, side and rear views, respectively, of the intervertebral implant of FIGS. **89-92** with the porous portions removed to show the support structure;

[0042] FIGS. **97** and **98** are top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. **99** is a cross-sectional view along the line **99-99** in FIG. **97**;

[0043] FIGS. **100** and **101** are side views of an intervertebral implant according to another embodiment of the disclosure, with FIG. **101** illustrating an insertion tool extending through the implant;

[0044] FIG. **102** is a perspective view of an intervertebral implant according to another embodiment of the disclosure, FIG. **103** is a cross-sectional view along the line **103-103** in FIG. **102** and FIG. **104** is a cross-sectional view along the line **104-104** in FIG. **102**;

[0045] FIGS. **105** and **106** are front and perspective views, respectively, illustrating a grid porous configuration;

[0046] FIG. **107** is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0047] FIG. **108** is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0048] FIG. **109** is a side view of an intervertebral implant according to another embodiment of the disclosure;

[0049] FIGS. **110-112** are perspective, side and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. **113** is a cross-sectional view along the line **113-113** in FIG. **112**;

[0050] FIGS. **114** and **115** illustrate alternative strut patterns of an illustrative support structure;

[0051] FIG. **116** is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0052] FIG. **117** is a cross-sectional view of an intervertebral implant according to another embodiment of the disclosure;

[0053] FIG. **118** is a top view of an intervertebral implant according to another embodiment of the

disclosure and FIG. 119 is a cross-sectional view along the line 119-119 in FIG. 118;
[0054] FIGS. 120 and 121 are top and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 122 is a cross-sectional view along the line 122-122 in FIG. 121;
[0055] FIG. 123 is a side view of an intervertebral implant according to another embodiment of the disclosure;
[0056] FIGS. 124 and 125 are top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 126 is a cross-sectional view along the line 126-126 in FIG. 124;
[0057] FIG. 127 is a side view of an intervertebral implant according to another embodiment of the disclosure and FIG. 128 is a cross-sectional view along the line 128-128 in FIG. 127;
[0058] FIG. 129 is an exploded side view of an intervertebral implant according to another embodiment of the disclosure;
[0059] FIGS. 130-132 are top views showing sequentially implantation of an expandable intervertebral implant according to another embodiment of the disclosure;
[0060] FIGS. 133 and 134 are front and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 135 is a cross-sectional view along the line 135-135 in FIG. 134;
[0061] FIG. 136 is a perspective view illustrating an example tool hole and FIG. 137 is a cross-sectional view illustrating a tool engaged in such a hole;
[0062] FIG. 138 is a schematic view of an example delivery tool in accordance with an embodiment of the disclosure;
[0063] FIGS. 139-141 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured; and
[0064] FIGS. 142 and 143 are perspective and top views, respectively, of the intervertebral implant of FIGS. 139-141 with the porous portions removed to show the support structure.

DETAILED DESCRIPTION

[0065] Embodiments of the disclosure are generally directed to intervertebral implants, systems, and method of use thereof. The implant may be suitable for use with the anterior, antero-lateral, lateral, and/or posterior portions of at least one motion segment unit of the spine. Traditionally, interbody spacers or implants intended to help facilitate intervertebral fusion may serve as a means to restore intervertebral height and/or lordosis. The implants may feature a central lumen to house bone graft material, for example. It is through this central lumen where most of the fusion may occur. The implants of the disclosure may incorporate a volumetric, interconnected porosity throughout the entire spacer or a portion thereof. This enables bone to grow into and/or through the spacer or a portion thereof, making it part of the fusion mass. The incorporation of a volumetric, interconnected porosity may encourage faster, stronger intervertebral fusion, thereby providing for better patient outcomes.

[0066] Various forms of additive manufacturing, or 3D printing, have been developed which allow structures to be formed layer by layer. One illustrative 3D printing technology is Direct Metal Laser Sintering (DMLS) wherein parts are built using a laser to selectively sinter (heat and fuse) a powdered metal material into layers. The process begins once a 3D CAD file is mathematically sliced into multiple 2D cross sections and uploaded into the system. After the first layer is produced, the build platform is lowered, another powder layer is spread across the plate, and the laser sinters the second layer. This process is repeated until the part is complete. Layer-by-layer manufacturing allows for the direct fabrication of complex parts that would be cost-prohibitive, and often impossible, to produce through traditional manufacturing processes. The powder layer thickness used during the fabrication of the spacers may be as thin as 30 μm . The resolution of the laser may be as fine as 70 μm . Although it is envisioned that any suitable thickness or laser resolution may be used or selected.

[0067] The disclosure is not limited to DMLS, but various 3D printing methods may be utilized. For example, VAT Photopolymerization utilizes a vat of liquid photopolymer resin which is cured through selective exposure to light (via a laser or projector) which then initiates polymerization and converts the exposed areas to a solid part. As another example, Powder Bed Fusion, of which DMLS is a subcategory, utilizes powdered materials which are selectively consolidated by melting it together using a heat source such as a laser or electron beam. The powder surrounding the consolidated part acts as support material for overhanging features.

[0068] As yet another example, in Binder Jetting Liquid bonding agents are selectively applied onto thin layers of powdered material to build up parts layer by layer. The binders include organic and inorganic materials. Metal or ceramic powdered parts are typically fired in a furnace after they are printed. Material Jetting is another example of a 3D printing process which may be utilized wherein droplets of material are deposited layer by layer to make parts. Common varieties include jetting a photocurable resin and curing it with UV light, as well as jetting thermally molten materials that then solidify in ambient temperatures. As another example, in Sheet Lamination sheets of material are stacked and laminated together to form an object. The lamination method can be adhesives or chemical (paper/plastics), ultrasonic welding, or brazing (metals). Unneeded regions are cut out layer by layer and removed after the object is built. Another example of a 3D printing process that may be utilized is Material Extrusion wherein material is extruded through a nozzle or orifice in tracks or beads, which are then combined into multi-layer models. Common varieties include heated thermoplastic extrusion and syringe dispensing. Yet another example is Directed Energy Deposition wherein powder or wire is fed into a melt pool which has been generated on the surface of the part where it adheres to the underlying part or layers by using an energy source such as a laser or electron beam.

[0069] The implants of the disclosure may be manufactured from any of these or other additive manufacturing processes currently known or later developed. The implants may also be manufactured utilizing a combination of additive manufacturing processes and other manufacturing processes, for example, laser etching. Additionally, the implants may be further processed during and/or after manufacture utilizing various techniques, for example, abrasion, machining, polishing, or chemical treatment. The implants may be manufactured from various materials, such as biocompatible materials, including metals, polymers, ceramics or combinations thereof. Exemplary materials include Titanium (and Titanium alloys), Cobalt-Chrome, PEEK, and/or Stainless Steel, for example.

[0070] As will be discussed in more detail hereinafter, the implants of the disclosure generally comprise a solid support structure and a porous structure formed integral therewith. The solid support structure may include solid front and rear walls interconnected by upper and lower implant surfaces. The upper and lower surfaces may include spaced apart rims with cross struts interconnecting the rims. In many embodiments, the solid support structure of the upper and lower surfaces includes a plurality of openings in which the integral porous structure is formed such that the porous structure extends along at least a portion of the upper and lower implant surfaces. The side walls extending between the front and rear walls generally have a minimal solid structure, for example, a plurality of struts extending between the upper and lower rims, but otherwise have open area therebetween in which the integral porous structure is formed. The configuration of the solid structure is selected to provide the implant sufficient structural integrity and mechanical stability while maximizing the area of porous structure which facilitates better integration/incorporation with the adjacent bone. In several embodiments of the disclosure, the solid structure generally encases the corners of the porous structure or otherwise houses the porous structure therein to maintain the structural integrity of the porous structure.

[0071] Referring now to FIGS. 1-7, one embodiment of a cervical intervertebral implant **10** will be described. As illustrated, the implant **10** has a body **11** with a generally trapezoidal shape. The body **11** is defined by a tapered front end **12**, a rectangular rear end **14** and side walls **16** and **18**

extending therebetween. The implant **10** has an outer perimeter OP extending about the body **11**. A hollow interior chamber **13** is defined within an inner perimeter IP of the body **11**. The hollow interior chamber **13** is configured to receive bone growth promoting materials, for example. The implant **10** has an upper surface **20** and a lower surface **22**, with both surfaces having a tapering portion **23** at the front end **12**. The upper and lower surfaces **20**, **22** may be substantially parallel or otherwise configured to provide the proper intervertebral spacing. The upper and lower surfaces **20**, **22** define a plurality of serrations **24** along the side walls **16**, **18** and a plurality of serrations **26** along the rear end **14**. The serrations **24**, **26** are defined by both the solid support structure **30** and the porous structure **50**. As will be described in detail hereinafter, the solid support structure **30** includes spaced apart rims **32**, **34** and **36**, **38** with cross struts **31** and **37**. The solid support structure **30** defines open spaces or recesses adjacent the cross struts **31**, **37** and the porous structure **50** is formed within such open spaces such that the solid structure **30** and the porous structure **50** together form the serrations **24**, **26**. As illustrated in FIGS. **1-4**, the porous structure **50** extends to and forms a portion of the implant upper and lower surfaces **20-22**. The rear end **14** of the implant **10** includes a hole **25** and a pair of blind slots **27** for receiving an instrument that is used for inserting the implant **10**. As seen in FIGS. **1-4**, the implant **10** is defined by a solid support structure **30** with an interfiled, integral porous structure **50**.

[0072] The solid support structure **30** will be described in more detail with reference to FIGS. **5-7**. An outer rim **32** extends about the outer perimeter OP of the upper surface **20** and an inner rim **34** extends about the inner perimeter IP of the upper surface **20**, i.e. about the interior chamber **13**. Similarly, an outer rim **36** extends about the outer perimeter OP of the lower surface **22** and an inner rim **38** extends about the inner perimeter IP of the lower surface **22**. A plurality of cross struts **31** extend between the outer rims **32**, **36** and the respective inner rims **34**, **38** along the side wall areas. As seen in the figures, the cross struts **31** along with contoured portions **33** of the rims **32**, **34**, **36**, **38** define the contour of the serrations **24**. In addition to interconnecting the rims within a given upper or lower surface, struts **44**, **46** and **48** extend within each side wall area to interconnect the upper rims **32**, **34** with the lower rims **36**, **38**. In the illustrated embodiment, a first strut **44** extends from the lower inner rim **38** to the upper outer rim **32** near the rear portion of the support structure **30**, a second strut **46** extends from the lower inner rim **38** to the upper outer rim **32** near the front portion of the support structure **30** and an X-shaped strut **48** extends between both lower rims **36**, **38** and both upper rims **32**, **34** at a central location of the support structure **30**. As can be seen in FIGS. **6** and **7**, each of the first struts **44** extends from the lower inner rim **38** proximate the rear wall **35** at an angle to approximately the midpoint of the upper outer rim **32**, substantially tangent to the curvature of the inner rims **34**, **38**. Similarly, each of the second struts **46** extends from the lower inner rim **38** proximate the front wall **40** at an angle to approximately the midpoint of the upper outer rim **32**, substantially tangent to the curvature of the inner rims **34**, **38**. Each of the X-shaped struts **48** extends substantially parallel to the upper and lower rims and positioned at the point where the first and second struts **44**, **46** meet with the upper outer rim **36**. The struts may have other configurations and more or fewer struts may be utilized.

[0073] The solid rear wall **35** additionally interconnects the outer rims **32**, **36** and the respective inner rims **34**, **38** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **35** defines the hole **25** and slots **27**. Recessed areas **39** and **41** on the upper and lower sides of the rear wall **35** define receiving areas for porous structure, as seen in FIGS. **1-4**. Cross members **37** in this area along with contours of the outer rims **32**, **36** define the serrations **26**. The solid front wall **40** has a concave configuration and also interconnects the outer rims **32**, **36** and the respective inner rims **34**, **38** along the front end area. The front wall **40** includes an upper sloped portion **42** extending between the upper outer rim **32** and inner rim **34** and a lower sloped portion **43** extending between the lower outer rim **36** and inner rim **38**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction

between the elements.

[0074] In the illustrations of the support structure **30** shown in FIGS. 5-7 with the porosity omitted for illustration, it is seen that there is significant open space between the upper rims **32, 34** and the lower rims **36, 38** with only the struts **44, 46, 48** therebetween. The struts **44, 46, 48** occupy only a minimal space between the upper and lower rims **32, 34, 36, 38**, for example, less than 50% of the wall space, thereby leaving substantial open space for the porous structure **50**. Additionally, there is open space between the inside surface of the front wall **40** and the inner rims **34, 38**. Furthermore, there is open space on an inside surface and the recesses **39, 41** of the rear wall **35**. As illustrated in FIGS. 1-4, in the implant **10**, these open spaces are filled with the porous structure **50** such that the porous structure **50** encapsulates the struts **44, 46, 48** and extends from the upper surface **20** to the lower surface **22** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **50** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **16, 18** along the outer perimeter OP.

[0075] The configuration of the support structure **30** and the porous structure **50** are selected, for example, to provide the implant with an adequate construct strength while maximizing the potential for bony in-growth and allowing for clear radiographic imaging. Referring to FIGS. 87 and 88, the porous structure **50** may have a randomized pattern of open pores **50a** or a repeating pattern of open pores **50b**. The porous structure **50** may have a suitable porosity (open volume). For example, the porous structure **50** may be greater than 50% open, greater than 60% open, greater than 70% open, or approximately 70% open, or approximately 75% open. The porous structure **50** may feature interconnected pores or open pores. The porous structure **50** may have pores, for example, ranging from approximately 100 μm -2 mm, approximately 100 μm -1 mm, approximately 200-900 μm , or approximately 300-800 μm in diameter. The pore size may have an average pore size of about 300-800 μm , about 400-700 μm , or about 500-600 μm . The pore size distribution may be unimodal or bi-modal. Although spherical or partially-spherical pores or nodes are exemplified in forming the porous structure, it is envisioned that other suitable pore shapes and configurations may be used, for example, repeating or random patterns of cylinders, cubes, cones, pyramids, polyhedrons, or the like.

[0076] It is contemplated that different areas of the support structure **30** may have varying stiffness or strength, for example, variable A-P stiffness to achieve optimized load on an anterior graft or to achieve a desired level of flexibility within the implant **10**. Furthermore, the porous structure **50** may have different porosities or densities in different areas of the implant **10**. For example, the porous structure **50** may have a higher porosity or density along the inner perimeter compared to that at the outer perimeter, for example, with the inner area having a cancellous porosity and the outer area having a cortical porosity. The porous structure **50** may have various configurations, for example, a grid or honeycomb pattern which may promote bony in-growth. Additionally, the porous structure **50** may be configured such that when it is turned past a critical angle it may appear opaque, thereby helping with assessment of the implant orientation or positioning. The surface texture of both the support structure and the porous structure may be controlled to provide both macro and micro texturizing. The features and characteristics described with respect to this embodiment may be incorporated in any of the embodiments described herein. Additionally, features described in any of the embodiments herein may be incorporated into any of the other embodiments.

[0077] Referring now to FIGS. 8-10, a cervical intervertebral implant **10'** in accordance with another embodiment of the disclosure will be described. The implant **10'** is similar to the previous embodiment except for a slight modification in the structure of the support structure **30'** and a corresponding modification in the porous structure **50'**. Compared to the previous embodiment, the rear wall **35'** has a narrower width with a portion of the rear end **14'** having an open support structure into which the porous structure **50'** extends. With the narrower width, the recesses portions **39', 41'** open directly into the open space of the side walls **16', 18'** and rear end **14'**. To

maintain sufficient implant strength, a pair of X-shaped struts **48'**, **48''** are positioned in each of the side wall areas **16'**, **18'** proximate the rear end **14'** of the implant **10'**. While the front end **12** of the implant **10'** remains substantially the same as in the previous embodiment, an additional X-shaped strut **48'''** is positioned in each of the side wall areas **16'**, **18'** proximate the rear end **14'** of the implant **10'**. Again, in the implant **10'**, the open spaces are filled with the porous structure **50'** such that the porous structure **50'** encapsulates the struts **44**, **46**, **48**, **48'**, **48''**, **48'''** and extends from the upper surface **20** to the lower surface **22** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **50'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **16'**, **18'** and a portion of the rear end **14'** along the outer perimeter OP.

[0078] Referring now to FIGS. **11-13**, a cervical intervertebral implant **10''** in accordance with another embodiment of the disclosure will be described. The implant **10''** is similar to the previous embodiment except for slight modification in the structure of the support structure **30''** and a corresponding modification in the porous structure **50''**. In the present embodiment, the struts within the side walls are replaced with external X-shaped struts **60**, **62**. Outer X-shaped struts **60** extend along each of the side walls **16''**, **18''** along the outer perimeter OP. The outer X-shaped struts **60** extend between the upper and lower outer rims **32** and **36**. Inner X-shaped struts **62** extend along each of the side walls **16''**, **18''** along the inner perimeter IP. The inner X-shaped struts **62** extend between the upper and lower inner rims **34** and **38**. A generally hollow wall space is defined between the outer and inner X-shaped struts **60**, **62** on the sides and the cross struts **31**, **37** on the top and bottom. These hollow wall spaces extend from the front wall **40** to the rear wall **35'** and are filled with the integral porous structure **50''**. Again, in the implant **10''**, the open spaces are filled with the porous structure **50''** such that it extends from the upper surface **20** to the lower surface **22** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment, the struts **60**, **62** are not encapsulated in the porous structure **50''**, but instead the struts **60** are coplanar with the porous structure **50''** along the outer perimeter OP and the struts **62** are coplanar with the porous structure **50''** along the inner perimeter IP. Again, the porous structure **50''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **16''**, **18''** and a portion of the rear end **14'** along the outer perimeter OP.

[0079] Referring now to FIGS. **14-16**, a cervical intervertebral implant **10'''** in accordance with another embodiment of the disclosure will be described. The implant **10'''** is similar to the previous embodiment except for slight modification in the structure of the support structure **30''** and a corresponding modification in the porous structure **50'''**. In the present embodiment, the external X-shaped struts **60'**, **62'** have a narrower configuration and have curved portions compared to those of the previous embodiment. Again, outer X-shaped struts **60'** extend along each of the side walls **16''**, **18''** along the outer perimeter OP as they extend between the upper and lower outer rims **32** and **36**. Inner X-shaped struts **62'** extend along each of the side walls **16''**, **18''** along the inner perimeter IP as they extend between the upper and lower inner rims **34** and **38**. In the present embodiment, in the rear area **14''** of the implant **10'''**, the rear wall **35''** is not connected to the upper or lower rim **32**, **36** and instead open spaces **61** extend therebetween. As in the previous embodiment, a generally hollow wall space is defined between the outer and inner X-shaped struts **60'**, **62'** on the sides and the cross struts **31**, **37** on the top and bottom. These hollow wall spaces extend from the front wall **40** to the rear wall **35''** and are filled with the integral porous structure **50'''**. As in the previous embodiments, all of the open spaces of the implant **10'''** are filled with the porous structure **50'''** such that it extends from the upper surface **20** to the lower surface **22** and from the outer perimeter OP to the inner perimeter IP. As in the previous embodiment, the struts **60'**, **62'** are not encapsulated in the porous structure **50'''**, but instead the struts **60'** are coplanar with the porous structure **50'''** along the outer perimeter OP and the struts **62'** are coplanar with the porous structure **50'''** along the inner perimeter IP. Again, the porous structure **50'''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **16''**, **18''** and a portion of the rear

end **14"** along the outer perimeter **OP**.

[0080] Referring now to FIGS. **17-19**, a cervical intervertebral implant **10.sup.iv** in accordance with another embodiment of the disclosure will be described. The implant **10.sup.iv** is similar to the previous embodiment except for slight modification in the structure of the support structure **30.sup.iv** and a corresponding modification in the porous structure **50.sup.iv**. In the present embodiment, the struts are replaced with an internal corrugated wall **64** within each of the side walls **16"**, **18"**. Each corrugated wall **64** extends between the upper support structure and the lower support structure. In the illustrated embodiment, each corrugated wall **64** extends from the front wall **12**, interconnects with the cross struts **31**, **37** and interconnects with the rear wall **35"**. In the present embodiment, in the rear area **14"** of the implant **10"**, open spaces **61** extend between the rear wall **35"** and the upper or lower rims **32**, **36** as in the previous embodiment, however, additional supports **63** extend between the upper rims **32**, **34** and the rear wall **35"** and between the lower rims **36**, **38** and the rear wall **35"**. As in the previous embodiments, all of the open spaces of the implant **10.sup.iv** are filled with the porous structure **50.sup.iv** such that it extends from the upper surface **20** to the lower surface **22**. The porous structure **50.sup.iv** of the present embodiment encapsulates each corrugated wall **64** and while the porous structure **50.sup.iv** is not continuous from the outer perimeter **OP** to the inner perimeter **IP**, the porous structure **50.sup.iv** still substantially defines the inner perimeter **IP** and defines a substantial portion of the side walls **16"**, **18"** and a portion of the rear end **14"** along the outer perimeter **OP**.

[0081] Referring now to FIGS. **20-27**, one embodiment of an anterior lumbar interbody fusion (ALIF) implant **110** will be described. As illustrated, the implant **110** has a body **111** with a generally D-shaped configuration. The body **111** is defined by a tapered front end **112**, a rectangular rear end **114** and side walls **116** and **118** extending therebetween. The implant **110** has an outer perimeter **OP** extending about the body **111**. A hollow interior chamber **113** is defined within an inner perimeter **IP** of the body **111**. The hollow interior chamber **113** is configured to receive bone growth promoting materials. The implant **110** has an upper surface **120** and a substantially parallel lower surface **122**, with both surfaces having a tapering portion **123** at the front end **112**. The upper and lower surfaces **120**, **122** define a plurality of serrations **124** along the side walls **116**, **118** and a plurality of serrations **126** along the rear end **114**. The rear end **114** of the implant **110** includes a plurality of screw holes **125** through which screws (not shown) extend to anchor the implant onto the vertebral body. Secondary holes **127** are provided to receive respective blocking set screws (not shown). A threaded hole **128** and a blind slot **129** are provided for receiving an instrument that is used for inserting the implant **110**. As seen in FIGS. **20-24**, the implant **110** is defined by a solid support structure **130** with an interfiled, integral porous structure **150**.

[0082] The solid support structure **130** will be described in more detail with reference to FIGS. **25-25**. An outer rim **132** extends about the outer perimeter **OP** of the upper surface **120** and an inner rim **134** extends about the inner perimeter **IP** of the upper surface **120**, i.e. about the interior chamber **113**. Similarly, an outer rim **136** extends about the outer perimeter **OP** of the lower surface **122** and an inner rim **138** extends about the inner perimeter **IP** of the lower surface **122**. A plurality of cross struts **131** extend between the outer rims **132**, **136** and the respective inner rims **134**, **138** along the side wall areas. As seen in the figures, the cross struts **131** along with contoured portions **133** of the rims **132**, **134**, **136**, **138** define the contour of the serrations **124**. In addition to interconnecting the rims within a given upper or lower surface, struts **144**, **146** extend within each side wall area to interconnect the upper rims **132**, **134** with the lower rims **136**, **138**. In the illustrated embodiment, a first multi-leg strut **144** extends from the lower inner rim **138** to the upper outer rim **132** near the rear portion of the support structure **130** and a second multi-leg strut **146** extends from the lower inner rim **138** to the upper outer rim **132** near the front portion of the support structure **130**.

[0083] A solid rear wall **135** additionally interconnects the outer rims **132**, **136** and the respective

inner rims **134**, **138** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **135** defines the holes **125**, **127**, **128** and the slot **129**. Recessed areas **139** on the upper and lower sides of the rear wall **135** define receiving areas for porous structure, as seen in FIGS. **20-24**. Cross members **137** in this area along with contours of the outer rims **132**, **136** define the serrations **126**. A solid front wall **140** with a concave configuration also interconnects the outer rims **132**, **136** and the respective inner rims **134**, **138** along the front end area. The front wall **140** includes an upper sloped portion **142** extending between the upper outer rim **132** and inner rim **134** and a lower sloped portion **143** extending between the lower outer rim **136** and inner rim **138**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0084] In the illustrations of the support structure **130** in FIGS. **25-27**, it is seen that there is significant open space between the upper rims **132**, **134** and the lower rims **136**, **138** with only the struts **144**, **146** therebetween. Additionally, there is open space between the inside surface of the front wall **140** and the inner rims **134**, **138**. Furthermore, there is open space on an inside surface and the recesses **139**, **141** of the rear wall **135**. As illustrated in FIGS. **20-24**, in the implant **110**, these open spaces are filled with the porous structure **150** such that the porous structure **150** encapsulates the struts **144**, **146** and extends from the upper surface **120** to the lower surface **122** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116**, **118** along the outer perimeter OP.

[0085] Referring now to FIGS. **28-34**, an ALIF implant **110'** in accordance with another embodiment of the disclosure will be described. The implant **110'** is similar to the previous embodiment except for slight modification in the structure of the support structure **130'** and a corresponding modification in the porous structure **150'**. Compared to the previous embodiment, the upper and lower surfaces **120'**, **122'** of the present implant **110'** are angled relative to one another. Additionally, the rear wall **135'** has a narrower width with a portion of the rear end **114'** having an open support structure into which the porous structure **150'** extends. With the narrower width, the recess portions **139'** open directly into the open space of the side walls **116'**, **118'** and rear end **114'**. The rear wall **135'** defines a single opening **128'** for receipt of an insertion tool. A cylinder **145** is positioned between the upper rims **132**, **136** and the lower rims **134**, **138** along the rear end **114'**. The cylinder **145** defines a through bore **147** configured to also receive an insertion tool. To maintain sufficient implant strength in the rear end **114'**, a first X-shaped struts **148** extends between the cylinder **145** and the end wall **135'** and a second X-shaped strut **148'** is positioned on the opposite side of the rear wall **135**. The front end **112** of the implant **110'** includes a recessed area **141** which defines a forward serration **149**. Again, in the implant **110'**, the open spaces are filled with the porous structure **150'** such that the porous structure **150'** encapsulates the struts **144**, **146**, **114**, **148'** and extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** and a portion of the rear end **114'** along the outer perimeter OP.

[0086] Referring now to FIGS. **35** and **36**, an ALIF implant **110''** in accordance with another embodiment of the disclosure will be described. The implant **110''** is similar to the previous embodiment except for slight modification in the structure of the support structure **130''** and a corresponding modification in the porous structure **150''**. Compared to the previous embodiment, the rear wall **135''** of the rear end **114''** includes a plurality of slots **129'** positioned about the hole **128'**. Additionally, the struts of the previous embodiment are replaced with a plurality of X-shaped struts **152** which are interconnected to one another by a circumferential intermediate rim **154**. Each of the struts **152** also interconnects with the upper rims **132**, **136** and the lower rims **134**, **137**. Again, in the implant **110''**, the open spaces are filled with the porous structure **150''** such that the

porous structure **150''** encapsulates the struts **152** and the intermediate rim **154** and extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** and a portion of the rear end **114''** along the outer perimeter OP.

[0087] Referring now to FIGS. **37** and **38**, an ALIF implant **110'''** in accordance with another embodiment of the disclosure will be described. The implant **110'''** is similar to the previous embodiment except for slight modification in the structure of the support structure **130''** and a corresponding modification in the porous structure **150''**. Compared to the previous embodiment, the X-shaped struts are replaced by coil struts **156a** and **156b**. Coil strut **156a** of side wall area **118'** extends from the rear wall **135''** to the front wall **140''** and extends about the circumferential intermediate rim **154**. Similarly, coil strut **156b** of side wall area **116'** extends from the rear wall **135''** to the front wall **140''** and extends about the circumferential intermediate rim **154**, however, the cylinder **145** extends through and interconnects with the coil strut **154b**. Each of the struts **156a**, **156b** also interconnects with the upper rims **132**, **136** and the lower rims **134**, **137**. Again, in the implant **110'''**, the open spaces are filled with the porous structure **150'''** such that the porous structure **150'''** encapsulates the intermediate rim **154** and struts **156a**, **156b** and extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** and a portion of the rear end **114''** along the outer perimeter OP.

[0088] Referring now to FIG. **39**, an ALIF implant **110.sup.iv** in accordance with another embodiment of the disclosure will be described. The implant **110.sup.iv** has a body **111'** with a generally oval configuration. The body **111'** is defined by a tapered front end **112''**, a rectangular rear end **114'''** and side walls **116''** and **118''** extending therebetween. The implant **110.sup.iv** has an outer perimeter OP extending about the body **111'**. A hollow interior chamber **113** is defined within an inner perimeter IP of the body **111'**. The hollow interior chamber **113** is configured to receive bone growth promoting materials. The implant **110.sup.iv** has an upper surface **120'** and a substantially parallel lower surface **122'**, with both surfaces having a tapering portion **123** at the front end **112**. In the present embodiment, each of the surfaces **120'**, **122'** includes a central surface portion **175**. The upper and lower surfaces **120'**, **122'** define a plurality of serrations **124'** along the side walls **116**, **118** and the central portions **175** and a plurality of serrations **126'** along the rear end **114'''**. The rear end **114'''** of the implant **110** includes a plurality of holes **128''**, **145** configured for receiving an instrument that is used for inserting the implant **110**. As in the previous embodiments, the implant **110.sup.iv** is defined by a solid support structure **130.sup.iv** with an interfiled, integral porous structure **150.sup.iv**.

[0089] The solid support structure **130.sup.iv** includes an upper plate **160** extending from the front end **112''** to the rear end **114'''** and defining side wall portions **162**, **164** and central portion **166**. A plurality of recesses **173** in the upper and lower plates **160**, **170** are filled with the porous structure **150.sup.iv** to define the serrations **124'**, **126'**. Similarly, a lower plate **170** extends from the front end **112''** to the rear end **114'''** and defines side wall portions **172**, **174** and central portion **176**. The upper and lower plates **160**, **170** are interconnected by a front wall **140''** and a rear wall **135''**. It is noted that in the present embodiment, the side walls **116'**, **118'** are generally open without any support structure and completely filled with the porous structure **150.sup.iv**. The rear wall **135''** defines the holes **128''**, **145**. The rear wall **135''** includes a plurality of recesses **171** configured to receive the porous structure **150.sup.iv**. As in the previous embodiments, the porous structure **150.sup.iv** generally extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150.sup.iv** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** along the outer perimeter OP.

[0090] Referring now to FIGS. 40-47, one embodiment of a transforaminal lumbar interbody fusion (TLIF) implant **210** will be described. As illustrated, the implant **210** has a body **211** with a generally rectangular shape. The body **211** is defined by a tapered front end **212**, a rectangular rear end **214** and side walls **216** and **218** extending therebetween. The implant **210** has an outer perimeter OP extending about the body **211**. A hollow interior chamber **213** is defined within an inner perimeter IP of the body **211**. The hollow interior chamber **213** is configured to receive bone growth promoting materials. The implant **210** has an upper surface **220** and a substantially parallel lower surface **222**, with both surfaces having a tapering portion **223** at the front end **212**. The upper and lower surfaces **220**, **222** define a plurality of serrations **224** along the side walls **216**, **218** and a plurality of serrations **226** along the rear end **214**. The rear end **214** of the implant **210** includes a hole **225** and a pair of slots **227** for receiving an instrument that is used for inserting the implant **210**. The implant **210** is defined by a solid support structure **230** with an interfiled, integral porous structure **250**.

[0091] The solid support structure **230** includes an outer rim **232** extending about the outer perimeter OP of the upper surface **220** and an inner rim **234** extending about the inner perimeter IP of the upper surface **220**, i.e. about the interior chamber **213**. Similarly, an outer rim **236** extends about the outer perimeter OP of the lower surface **222** and an inner rim **238** extends about the inner perimeter IP of the lower surface **222**. A plurality of cross struts **231** extend between the outer rims **232**, **236** and the respective inner rims **234**, **238** along the side wall areas. As seen in the figures, the cross struts **231** along with contoured portions **233** of the rims **232**, **234**, **236**, **238** define the contour of the serrations **224**. In addition to interconnecting the rims within a given upper or lower surface, external radial struts **260**, **262** additionally interconnect the rims **232**, **234**, **236**, **238**. Outer radial struts **260** extend along each of the side walls **216**, **218** along the outer perimeter OP. The outer radial struts **260** have a central portion **261** and legs **263** which extend between the upper and lower outer rims **232** and **236**. Inner radial struts **262** extend along each of the side walls **216**, **218** along the inner perimeter IP. The inner radial struts **262** have a central portion **265** and legs **267** which extend between the upper and lower inner rims **234** and **238**.

[0092] A solid rear wall **235** additionally interconnects the outer rims **232**, **236** and the respective inner rims **234**, **238** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **235** defines the hole **225** and slots **227**. Recessed areas **239** and **241** on the upper and lower sides of the rear wall **235** define receiving areas for porous structure, as seen in FIGS. 40-43. The contours of the outer rims **232**, **236** define the serrations **226**. A solid front wall **240** with a concave configuration also interconnects the outer rims **232**, **236** and the respective inner rims **234**, **238** along the front end area. The front wall **240** includes an upper sloped portion **242** extending between the upper outer rim **232** and inner rim **234** and a lower sloped portion **243** extending between the lower outer rim **236** and inner rim **238**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0093] In the illustrations of the support structure **230** in FIGS. 44-47, it is seen that there is significant open space between the upper rims **232**, **234** and the lower rims **236**, **238**, between the inside surface of the front wall **240** and the inner rims **234**, **238**, and on an inside surface and the recesses **239**, **241** of the rear wall **235**. As illustrated in FIGS. 40-43, in the implant **210**, the open spaces are filled with the porous structure **250** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment, the struts **260**, **262** are not encapsulated in the porous structure **250**, but instead the struts **260** are coplanar with the porous structure **250** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250** along the inner perimeter IP. Again, the porous structure **250** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0094] Referring now to FIGS. 48-51, a TLIF implant **210'** in accordance with another embodiment of the disclosure will be described. The implant **210'** is similar to the previous embodiment except for slight modification in the structure of the support structure **230'** and a corresponding modification in the porous structure **250'**. In the present embodiment, a connecting ring **268** interconnects the central portion **261** with the central portion **265** of the struts **260**, **262** in each side wall **216**, **218**. Additionally, on the upper and lower surfaces **220**, **222**, additional serrations **229** are provided adjacent the rear end **214'**. Again, in the implant **210'**, the open spaces are filled with the porous structure **250'** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment the struts **260** are coplanar with the porous structure **250'** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250'** along the inner perimeter IP. The connecting rings **268** are encapsulated within the porous structure **250'**. Again, the porous structure **250'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0095] Referring now to FIGS. 52 and 53, a TLIF implant **210''** in accordance with another embodiment of the disclosure will be described. The implant **210''** is similar to the embodiment illustrated in FIGS. 40-47 except for slight modification in the structure of the support structure **230''** and a corresponding modification in the porous structure **250''**. In the present embodiment, in addition to the struts **260**, **262**, a plurality of X-shaped struts **244** extend between the upper rims **232**, **234** and the lower rims **236**, **238** in each side wall **216**, **218**. Again, in the implant **210''**, the open spaces are filled with the porous structure **250''** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment the struts **260** are coplanar with the porous structure **250'** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250'** along the inner perimeter IP. The X-shaped struts **244** are encapsulated within the porous structure **250''**. Again, the porous structure **250''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0096] Referring now to FIGS. 52 and 53, a TLIF implant **210'''** in accordance with another embodiment of the disclosure will be described. The implant **210'''** is similar to the embodiment illustrated in FIGS. 40-47 except for slight modification in the structure of the support structure **230'''** and a corresponding modification in the porous structure **250'''**. In the present embodiment, the side walls **216'**, **218'** do not include external struts, but instead include a plurality of X-shaped struts **246** extending between the upper rims **232**, **234** and the lower rims **236**, **238** in each side wall **216'**, **218'**. Additionally, an intermediate plate **248** interconnects the X-shaped struts **246** within each side wall area. Again, in the implant **210'''**, the open spaces are filled with the porous structure **250'''** such that the porous structure **250'''** encapsulates the struts **246** and the intermediate plate **248** and extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **250'''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216'**, **218'** along the outer perimeter OP.

[0097] Referring now to FIGS. 58-64, one embodiment of a lateral lumbar interbody fusion (LLIF) implant **310** will be described. As illustrated, the implant **310** has a body **311** with a generally rectangular shape. The body **311** is defined by a tapered front end **312**, a rectangular rear end **314** and side walls **316** and **318** extending therebetween. The implant **310** has an outer perimeter OP extending about the body **311**. A hollow interior chamber **313** is defined within an inner perimeter IP of the body **311**. The hollow interior chamber **313** is configured to receive bone growth promoting materials. The implant **310** has an upper surface **320** and a substantially parallel lower surface **322**, with both surfaces having a tapering portion **323** at the front end **312**. The upper and lower surfaces **320**, **322** define a plurality of serrations **324** along the side walls **316**, **318**, a serration **328** along the front end **312** and a plurality of serrations **326** along the rear end **314**. The

illustrated serrations **324**, **326**, **328** have micro serrations defined thereon. The rear end **314** of the implant **310** includes a hole **325** surrounded by a slot **327** for receiving an instrument that is used for inserting the implant **310**. The implant **310** is defined by a solid support structure **330** with an interfiled, integral porous structure **350**.

[0098] Referring to FIGS. **62-64**, the solid support structure **330** includes an outer rim **332** extending about the outer perimeter OP of the upper surface **320** and an inner rim **334** extending about the inner perimeter IP of the upper surface **320**, i.e. about the interior chamber **313**. Similarly, an outer rim **336** extends about the outer perimeter OP of the lower surface **322** and an inner rim **338** extends about the inner perimeter IP of the lower surface **322**. A plurality of cross struts **331** extend between the outer rims **332**, **336** and the respective inner rims **334**, **338** along the side wall areas. As seen in the figures, the cross struts **331** along with contoured portions **333** of the rims **332**, **334**, **336**, **338** define the contour of the serrations **324**. In addition to interconnecting the rims within a given upper or lower surface, a plurality of X-shaped struts **344** extend within each side wall area to interconnect the upper rims **332**, **334** with the lower rims **336**, **338**.

[0099] A solid rear wall **335** additionally interconnects the outer rims **332**, **336** and the respective inner rims **334**, **338** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **335** defines the hole **325** and slot **327**. A portion of the rear wall **335** defining the hole **325** extends to a secondary rear wall **360** which extends between the upper and lower inner rims **334**, **338**. A cross strut **339** on the upper and lower sides of the rear wall **335** and a portion of the rear wall **335** define the serrations **326**. A solid front wall **340** also interconnects the outer rims **332**, **336** and a secondary front wall **362** interconnects the inner rims **334**, **338** along the front end area. The front wall **340** includes an upper sloped portion **342** extending between the upper outer rim **332** and inner rim **334** and a lower sloped portion **343** extending between the lower outer rim **336** and inner rim **338**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements. A portion of the front wall **340** defines the serration **328**.

[0100] In the illustrations of the support structure **330** in FIGS. **62-64**, it is seen that there is significant open space between the upper rims **332**, **334** and the lower rims **336**, **338**, between the inside surface of the front wall **340** and the secondary wall **362** and on an inside surface of the rear wall **335**. As illustrated in FIGS. **58-61**, in the implant **310**, the open spaces are filled with the porous structure **350** such that it encapsulates the struts **344** and extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **350** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316**, **318** along the outer perimeter OP.

[0101] Referring now to FIGS. **65-67**, an LLIF implant **310'** in accordance with another embodiment of the disclosure will be described. The implant **310'** is similar to the previous embodiment except for slight modification in the structure of the support structure **330'** and a corresponding modification in the porous structure **350'**. Compared to the previous embodiment, the cross struts **331'** and **339'** do not include mini-serrations. Additionally, the rear wall **335'** has a narrower width with a portion of the rear end **314'** having an open support structure into which the porous structure **350'** extends. The front end **312** of the implant **310'** includes a cylindrical portion **364** extending between the front wall **340** and the secondary front wall **362**. Again, in the implant **310'**, the open spaces are filled with the porous structure **350'** such that the porous structure **350'** encapsulates the struts **344** and extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **350'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316'**, **318'** along the outer perimeter OP.

[0102] Referring now to FIGS. **68** and **69**, an LLIF implant **310''** in accordance with another embodiment of the disclosure will be described. The implant **310''** is similar to the previous

embodiment except for slight modification in the structure of the support structure **330''** and a corresponding modification in the porous structure **350''**. Compared to the previous embodiment, the upper rims **332'**, **334'** and lower rims **336'**, **338'** each have a wider configuration and do not have cross struts extending therebetween. The serrations **324'**, **326'**, **328'** are formed directly on the rims **332'**, **334'**, **336'**, **338'**, on the rear wall **335'** and the front wall **340'**. Additionally, the struts **344** are interconnected to one another by a circumferential intermediate rim **365**. Again, in the implant **310''**, the open spaces are filled with the porous structure **350''** such that the porous structure **350''** encapsulates the struts **344** and intermediate rim **365** and extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **350''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316''**, **318''** along the outer perimeter OP.

[0103] Referring now to FIGS. **70-72**, an LLIF implant **310'''** in accordance with another embodiment of the disclosure will be described. The implant **310'''** is similar to the embodiment illustrated in FIGS. **65-67** except for slight modification in the structure of the support structure **330'''** and a corresponding modification in the porous structure **350''**. Compared to the embodiment illustrated in FIGS. **65-67**, the rims **332**, **334**, **336**, **338** do not have cross struts extending therebetween. Additionally, external struts **366** are provided along the outside perimeter OP along each side wall **316'''**, **318'''** and external struts **368** are provided along the inside perimeter IP along each side wall **316'''**, **318'''**. Again, in the implant **310'''**, the open spaces are filled with the porous structure **350'''** such that the porous structure **350'''** encapsulates the struts **344** and extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. The struts **366** are coplanar with the porous structure **350'''** along the outer perimeter OP and the struts **368** are coplanar with the porous structure **350'''** along the inner perimeter IP. In the illustrated embodiment, the porous structure **350'''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316'''**, **318'''** along the outer perimeter OP.

[0104] Referring now to FIGS. **73-75**, an LLIF implant **310.sup.iv** in accordance with another embodiment of the disclosure will be described. The implant **310.sup.iv** is similar to the previous embodiment except for slight modification in the structure of the support structure **330.sup.iv** and a corresponding modification in the porous structure **350.sup.iv**. Compared to the previous embodiment, the rims **332**, **334**, **336**, **338** have cross struts **331'** extending therebetween but do not include X-shaped struts within the side walls **316.sup.iv**, **318.sup.iv**. Again, in the implant **310.sup.iv**, the open spaces are filled with the porous structure **350.sup.iv** such that the porous structure **350.sup.iv** extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. The struts **366** are coplanar with the porous structure **350.sup.iv** along the outer perimeter OP and the struts **368** are coplanar with the porous structure **350.sup.iv** along the inner perimeter IP. In the illustrated embodiment, the porous structure **350.sup.iv** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316.sup.iv**, **318.sup.iv** along the outer perimeter OP.

[0105] Referring now to FIGS. **139-143**, an LLIF implant **310.sup.v** in accordance with another embodiment of the disclosure will be described. The implant **310.sup.v** is similar to the embodiment illustrated in FIGS. **58-64** except for slight modification in the structure of the support structure **330.sup.v** and a corresponding modification in the porous structure **350.sup.v**. In the present embodiment, the implant body **311'** has a wedge configuration, tapering from a thickest height along the outer edge of the side wall **318.sup.v** to a thinnest height along the outer edge of the side wall **316.sup.v**. The front wall **340''** and rear wall **335''** are correspondingly tapered. Additionally, the serration **328''** along the front wall **340''** does not include micro serrations. As in the embodiment illustrated in FIGS. **58-64**, the rims **332**, **334**, **336**, **338** have cross struts **331** extending therebetween, but do not include X-shaped struts within the side walls **316.sup.v**, **318.sup.v**. Instead, each side wall **316.sup.v**, **318.sup.v** has a single linear strut **369** extending perpendicularly between the upper outer rim **332** and the lower outer rim **336** at an approximate

mid-point of the side wall **316.sup.v**, **318.sup.v**. Again, in the implant **310.sup.v**, the open spaces are filled with the porous structure **350.sup.v** such that the porous structure **350.sup.v** extends from the upper surface **320** to the lower surface **322** and from the outer perimeter OP to the inner perimeter IP. The struts **369** are coplanar with the porous structure **350.sup.v** along the outer perimeter OP. In the illustrated embodiment, the porous structure **350.sup.v** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316.sup.v**, **318.sup.v** along the outer perimeter OP.

[0106] Referring now to FIGS. **76-86**, embodiments of two-piece cervical implants **400**, **400'** will be described. Each of the implants **400**, **400'** includes a plate **402**, **402'** and connectable spacer **410**. Each of the plates **402**, **402'** defines bone screw holes **403** and one or more blocking set screws **404**. The plates **402**, **402'** may have varying configurations with tabs **405** and projections **406**. The plate configurations are not limited to those shown. Each of the plates **402**, **402'** includes arms **407** with inward projections **408** which engage in slots **419** in the spacer **410** to connect the spacer **410** with the respective plate **402**, **402'**. The illustrated plates **402**, **402'** are solid structures and do not include porous structure, however, the plates may be made with portions having a porous structure.

[0107] Referring to FIGS. **80-86**, the spacer **410** has body **411** with a generally U-shaped configuration. The body **411** is defined by a tapered front end **412** with side walls **416** and **418** extending to free ends **414a**, **414b**. The spacer **410** has an outer perimeter OP extending about the body **411** and an inner perimeter IP within the U-shape of the body **411**. When the spacer **410** is connected with a plate **402**, **402'**, a hollow interior chamber **413** is defined which is configured to receive bone growth promoting materials. The spacer **410** has an upper surface **420** and a substantially parallel lower surface **422**, with both surfaces having a tapering portion **423** at the front end **412**. The upper and lower surfaces **420**, **422** define a plurality of serrations **424** along the side walls **416**, **418**, a serration **428** along the front end **412** and serrations **426** at the free ends **414a**, **414b**. Each side wall **416**, **418** defines a connection slot **419** forward of the respective free end **414a**, **414b**. The slots **419** are engaged by the projections **408** on the plate arms **407** to attach the spacer **410** to the respective plate **402**, **402'**. The spacer **410** is defined by a solid support structure **430** with an interfiled, integral porous structure **450**.

[0108] Referring to FIGS. **84-86**, the solid support structure **430** includes an outer rim **432** extending about the outer perimeter OP of the upper surface **420** and an inner rim **434** extending about the inner perimeter IP of the upper surface **420**. Similarly, an outer rim **436** extends about the outer perimeter OP of the lower surface **422** and an inner rim **438** extends about the inner perimeter IP of the lower surface **422**. A plurality of cross struts **431** extend between the outer rims **432**, **436** and the respective inner rims **434**, **438** along the side wall areas. As seen in the figures, the cross struts **431** along with contoured portions **433** of the rims **432**, **434**, **436**, **438** define the contour of the serrations **424**.

[0109] A solid front wall **440** interconnects the outer rims **432**, **436** along the front end area. The front wall **440** includes an upper sloped portion **442** and a lower sloped portion **443**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements. A portion of the front wall **440** defines the serration **428**. A rear wall structure **435a**, **435b** at each free end **414a**, **414b** additionally interconnects the outer rims **432**, **436** and the respective inner rims **434**, **438** along the rear end area as well as further connecting the upper and lower structures together. The rear wall structure **435a** extends about an open area **460** with a side opening **462** and rear wall structure **435b** extends about an open area **464** with opposed side openings **465**, **466**.

[0110] In the illustrations of the support structure **430** in FIGS. **84-86**, it is seen that there is significant open space between the upper rims **432**, **434** and the lower rims **436**, **438**, between the inside surface of the front wall **440** and the inner rims **434**, **438** and within the open areas **460**, **462**. As illustrated in FIGS. **80-83**, in the spacer **410**, the open spaces are filled with the porous structure

450 such that it extends from the upper surface 420 to the lower surface 422 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 450 substantially defines the inner perimeter IP and defines a substantial portion of the side walls 416, 418 along the outer perimeter OP.

[0111] Referring now to FIGS. 89-96, an embodiment of an articulating TLIF implant 510 will be described. As illustrated, the implant 510 has a body 511 with a generally arcuate shape. The body 511 is defined by a tapered front end 512, a rectangular rear end 514 and arcuate side walls 516 and 518 extending therebetween. The implant 510 has an outer perimeter OP extending about the body 511. A hollow interior chamber 513 is defined within an inner perimeter IP of the body 511. The hollow interior chamber 513 is configured to receive bone growth promoting materials. The implant 510 has an upper surface 520 and a substantially parallel lower surface 522, with both surfaces having a tapering portion 523 at the front end 512. The upper and lower surfaces 520, 522 define a plurality of serrations 524 along the side walls 516, 518 and a plurality of serrations 526 along the rear end 514. The rear end 514 of the implant 510 includes a hole 525 through one of the surfaces 522 and a slot 527 in communication therewith. An articulating member 570 is positioned within the hole 525 and slot 527 and is pivotal relative to the body 511. The articulating member 570 includes a body 572 received in and rotatable within the hole 525. A threaded tool receiving opening 574 extends from the body 572 and is aligned within the slot 527. A threaded implant tool is received in the threaded tool receiving opening 574 and may be utilized for implanting and articulating the position of the implant 510 in a known manner. The articulating member 570 may be manufactured during the 3D manufacturing process of the body 511 or may be manufactured separately and thereafter positioned within the body 511. The implant 510 is defined by a solid support structure 530 with an interfiled, integral porous structure 550.

[0112] The solid support structure 530 includes an outer rim 532 extending about the outer perimeter OP of the upper surface 520 and an inner rim 534 extending about the inner perimeter IP of the upper surface 520, i.e. about the interior chamber 513. Similarly, an outer rim 536 extends about the outer perimeter OP of the lower surface 522 and an inner rim 538 extends about the inner perimeter IP of the lower surface 522. A plurality of cross struts 531 extend between the outer rims 532, 536 and the respective inner rims 534, 538 along the side wall areas. As seen in the figures, the cross struts 531 along with contoured portions 533 of the rims 532, 534, 536, 538 define the contour of the serrations 524. In addition to interconnecting the rims within a given upper or lower surface, external linear struts 560, 562 additionally interconnect the rims 532, 534, 536, 538. An outer linear strut 560 extends along each of the side walls 516, 518 along the outer perimeter OP. The outer linear struts 560 extend substantially perpendicular to the upper and lower rims at an approximate midpoint of the respective wall. An inner linear strut 562 extends along each of the side walls 516, 518 along the inner perimeter IP. The inner linear struts 562 extend substantially perpendicular to the upper and lower rims at an approximate midpoint of the respective wall.

[0113] The solid rear wall 535 additionally interconnects the outer rims 532, 536 and the respective inner rims 534, 538 along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall 535 defines the hole 525 and slots 527. Recessed areas 539 and 541 on the upper and lower sides of the rear wall 535 extend between the cross struts 537 and define the serrations 526, however, in the present embodiment, these recesses generally do not receive porous structure, as seen in FIGS. 89-92. It is understood that these recesses could receive porous structure as in previous embodiments. The solid front wall 540 also interconnects the outer rims 532, 536 and the respective inner rims 534, 538 along the front end area. The front wall 540 includes an upper sloped portion 542 extending between the upper outer rim 532 and inner rim 534 and a lower sloped portion 543 extending between the lower outer rim 536 and inner rim 538. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0114] In the illustrations of the support structure **530** in FIGS. **93-96**, it is seen that there is significant open space between the upper rims **532**, **534** and the lower rims **536**, **538** and between the inside surface of the front wall **540** and the inner rims **534**, **538**. As illustrated in FIGS. **89-92**, in the implant **510**, the open spaces are filled with the porous structure **550** such that it extends from the upper surface **520** to the lower surface **522** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment, the struts **560**, **562** are not encapsulated in the porous structure **550**, but instead the struts **560** are coplanar with the porous structure **550** along the outer perimeter OP and the struts **562** are coplanar with the porous structure **550** along the inner perimeter IP. Again, the porous structure **550** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **516**, **518** along the outer perimeter OP.

[0115] Referring to FIGS. **97-99**, another embodiment of an intervertebral implant **600** will be described. The implant **600** includes an upper plate **602** and a lower plate **604** supported by a plurality of solid struts **606**, **608** extending therebetween. An open graft window **605** is defined within the implant **600**. The struts **606**, **608** are configured to provide desired load bearing properties of the implant **600**. The end plates **602**, **604** may be formed as a porous structure, for example, having a trabecular porosity. In addition to the porosity, the surfaces of the end plates **602**, **604** may have a nanoscale roughness formed thereon. As seen in FIG. **98**, in the illustrated embodiment, the end plates **602**, **604** and the struts **606**, **608** are configured such that the implant **600** has a biconvex superior and inferior geometry.

[0116] Referring to FIGS. **100** and **101**, another embodiment of an intervertebral implant **610** will be described. The implant **610** includes an upper plate **612** and a lower plate **614** supported by X-shaped struts **616** extending between the plates **612**, **614** on each lateral side of the implant **610**. An open graft window **615** is defined within the implant **610** between the plates **612**, **614**. As shown in FIG. **101**, the window **615** may be configured to receive an insertion tool **618**. The end plates **602**, **604** and struts **616** may be formed as a porous structure. Alternatively, the struts **616** may be formed as a solid structure. In addition to the porosity, the surfaces of the end plates **612**, **614** may have a roughness **613** formed thereon.

[0117] Referring to FIGS. **102-106**, another embodiment of an intervertebral implant **620** will be described. The implant **620** includes an upper plate **622** and a lower plate **624** supported by opposed solid walls **616**, **618** extending therebetween. Anterior and/or lateral windows **621** open into an interior chamber **623** configured to receive graft material. Each of the upper and lower plates **622**, **624** has a central open area **625** wherein a porous webbing surface **627** is formed. FIGS. **105** and **106** illustrate examples of a porous webbing surface **627**. The webbing may be formed with a grid or honeycomb pattern which is porous to bone ingrowth. The webbing may also be configured such that it may appear opaque when turned past a critical angle. Such a feature may be used for assessing the implants orientation or position.

[0118] Referring to FIGS. **107** and **108**, intervertebral implants **630**, **640** illustrating various features will be described. The implant **630** of FIG. **107** includes a body **632** with an upper surface **634** and a lower surface **636**. One or both of the surfaces **634**, **636** may be formed with a complex configuration to optimize end plate contact. For example, the upper surface **634** has a biconcave configuration. Similarly, the implant **640** of FIG. **108** includes a body **642** with an upper surface **644** and a lower surface **646**. One or both of the surfaces **644**, **646** may be formed with various teeth or surface roughness features to minimize subsidence or migration. For example, the upper surface **646** has a plurality of teeth **648** defined thereon. For each of the implants **630**, **640**, the body **632**, **642** may include both solid structure and porous structure as described herein.

[0119] Referring to FIG. **109**, another embodiment of an intervertebral implant **650** will be described. The implant **650** includes an upper plate **652** and a lower plate **654** supported by a plurality of solid struts **656** extending therebetween. The density of the struts **656** increases moving posteriorly to provide a varying A-P stiffness to the implant, for example, to optimize the load of an anterior graft.

[0120] Referring to FIGS. **110-113**, another embodiment of an intervertebral implant **660** will be described. The implant **660** includes an upper surface **662** and a lower surface **664** supported by a plurality of solid struts **666** extending therebetween. The upper and lower surfaces **662**, **664** are defined by perimeter struts **667** and diagonal struts **668**. As illustrated, the struts **666**, **667**, **668** that make up the support structure extend generally along the supporting edges while the remainder of the implant structure remains open for either porous structure or ingrowth chambers. FIGS. **114** and **115** illustrate alternative strut configurations for either the side walls or top or bottom surfaces of the implant **650**. In the embodiment illustrated in FIG. **114**, the struts **666'**, **668'** have a rectangular central region with radial supports extending from the corners thereof. In the embodiment illustrated in FIG. **115**, the struts **666''**, **668''** have a diamond shaped central region with octagon patterned struts thereabout.

[0121] Referring to FIG. **116**, another embodiment of an intervertebral implant **670** will be described. The implant **670** includes a body **671** with an upper surface **672** and a lower surface **672** extending about an opening **613** into an ingrowth cavity. The front end of the implant **670** includes sloped surfaces **675** to define a smooth leading edge. One or both of the surfaces **672**, **672** may include a radial sawtooth configuration **676**. As in previous embodiments, the body **671** may include both solid structure and porous structure as described herein.

[0122] Referring to FIG. **117**, another embodiment of an intervertebral implant **680** will be described. The implant **680** includes a solid support frame **682** encased within porous structure **684**, **686**. The implant **680** defines at least one opening **681** into an ingrowth chamber **683**. In the illustrated embodiment, the porous structure includes areas of different densities. In the illustrated embodiment, the outer porous structure **684** has a lower density than the inner porous structure **686**. As described herein, the various implants may be formed with porous structures having varying densities, not just with respect to outer and inner layers, but also within a given layer at different areas of the implant, e.g. anterior versus posterior.

[0123] Referring to FIGS. **118** and **119**, another embodiment of an intervertebral implant **690** will be described. The implant **690** includes a body **692** extending about an ingrowth chamber **693**. The body **692** is defined by a porous portion **696** surrounding a plurality of support struts **694**. The support struts **694** are not interconnected to one another. The support struts **694** may be a solid structure or may be a porous structure with a greater density than the porous portion **696**. For example, the struts **694** may be a porous structure having a cancellous density while the porous structure **696** has a cortical density. Since the body **692** includes a significant porous structure, the ingrowth chamber **693** may be smaller than compared to an implant having a non-porous body.

[0124] Referring to FIGS. **120-122**, another embodiment of an intervertebral implant **700** will be described. The implant **700** includes a body **702** with an ingrowth pocket **703** extending into the posterior side thereof. The body **702** is defined by a porous portion **706** surrounding a plurality of support struts **704**. The support struts **704** are not interconnected to one another. The support struts **704** may be a solid structure or may be a porous structure with a greater density than the porous portion **706**. For example, the struts **704** may be a porous structure having a cancellous density while the porous structure **706** has a cortical density.

[0125] Referring to FIG. **123**, another embodiment of an intervertebral implant **710** will be described. The implant **710** includes a body **712**. The body **712** is defined by an external area of higher density porous structure **714** surrounding a less dense porous structure **716**. The higher density porous structure **714** acts as struts to provide the implant **710** strength. The area of higher density porous structure **714** is shown in a zig zag pattern, but may have other patterns and configurations.

[0126] Referring to FIGS. **124-126**, another embodiment of an intervertebral implant **720** will be described. The implant **720** includes an upper plate **722** and a lower plate **724**. Each of the plates **722** and **724** has a wave configuration with minimum struts **726** interconnecting the plates **722**, **724**. With such a configuration, the implant **720** has spring between the plates. An open graft

window **723** is defined within the implant **720**. The plates **722**, **724** and the struts **726** may include both solid structure and porous structure, as described herein.

[0127] Referring to FIGS. **127** and **128**, another embodiment of an intervertebral implant **730** will be described. The implant **730** includes a body **732** surrounding an ingrowth chamber **733**. The body **732** is defined by a porous portion **734** which is surrounded a continuously wrapped support rib **736**. The support rib **736** may be a solid structure or may be a porous structure with a greater density than the porous portion **734**.

[0128] Referring to FIG. **129**, another embodiment of an intervertebral implant **740** will be described. The implant **740** includes interconnectable body portions **742** and **744**. In the illustrated embodiment, the upper body portion **742** includes a post **743** configured to friction fit within a hole **745** of the lower body portion **744**. Various interconnectable upper and lower body portions **742**, **744** may be printed and interconnected to form a customized implant **740**. In the illustrated embodiment, three additional lower body portions **744'**, **744''**, **744'''** and provided and all may be interconnected with the upper body portion **742** to achieve implants having different configurations.

[0129] Referring to FIGS. **130-132**, another embodiment of an intervertebral implant **750** and a method of implantation thereof will be described. The implant **750** includes a central body **752** and a pair of expandable wings **753**, **755**. The implant **750** is connected to a hollow insertion tool **756** and moved within an intervertebral space **759**. Once in position, graft material **757** is fed through the tool **756** and into the implant **750** whereby the wings **753**, **755** are caused to expand. Graft material **757** is supplied until the area within the central body **752** and the expanded wings **753**, **755** is filled. The packed graft material **757** maintains the expanded configuration of the implant **750**.

[0130] Referring to FIGS. **133-135**, another embodiment of an intervertebral implant **760** and a method of inserting such implant will be described. The implant **760** includes a body **762** extending about an ingrowth chamber **763**. The body **762** is generally defined by a porous structure **764**. Solid bore holes **766** are defined within the body **762** and are configured to receive screws or other anchors to facilitate fixation of the implant **760**. The body **762** also defines a tool receiving opening **765**. In the illustrated embodiment, an expandable tool **770** is inserted through the opening **765**. The expandable tool **770** includes a pair of legs **772** with flanges **773** on the end thereof. In a collapsed condition, the flanges **773** pass through the opening **765**. To engage the implant **760**, a needle **775** or the like is extended between the legs **772**, thereby forcing the flanges **773** outward such that the engaged the inside surface of the implant and do not pass through the opening **765** until the needle **775** is removed.

[0131] Referring to FIGS. **136** and **137**, another tool and implant opening interconnection assembly will be described. In the present embodiment, the implant **780** has a tool receiving opening **782** with a wider central region **783** and narrower end regions **784**. The insertion tool **790** has a head **792** with a complimentary configuration. The head **792** is positioned within the opening **782** with the orientation of the head **792** matching that of the opening **782** such that the head **792** passes through the opening **782**. Once inside, the tool head **792** is rotated, for example, a quarter turn, such that the tool head **792** engages the inside surface of the implant **780** and is no longer removable through the opening **782**. A securing sleeve **794** is threadably advanced along the tool **790** and engages the outside surface of the implant **780**, securing the implant **780** between the tool head **792** and the sleeve **794**.

[0132] Referring to FIG. **138**, another tool and implant opening interconnection assembly will be described. In the present embodiment, the implant **800** has a tool receiving opening **802** with an outward taper **803** at the inner side of the opening **802**. The insertion tool **810** has an outer hollow tube **812** with a collet **814** on the free end thereof and an inner hollow tube **816** with a through passage **817**. The inner hollow tube **816** is axially moveable relative to the outer hollow tube **812**. Initially, the inner hollow tube **816** is withdrawn into the outer hollow tube **812** such that it is clear

of the collet **814** such that the collet **814** may collapse and pass through the implant opening **802**. Once the collet **814** passes through the opening **802**, the inner hollow tube **816** is moved forward such that the collet **814** is expanded outward and maintained in the outward position, engaging the tapered portion **803** of the implant opening **802**. Graft material or the like may be passed into the implant **800** through the through passage **817**.

[0133] As described herein, the implants of the disclosure generally comprise a solid or higher density support structure and a porous structure formed integral therewith. The solid support structure may include solid front and rear walls interconnected by upper and lower implant surfaces. In several of the embodiments, the upper and lower surfaces include spaced apart rims with cross struts interconnecting the rims. In many embodiments, the solid support structure of the upper and lower surfaces includes a plurality of openings in which the integral porous structure is formed such that the porous structure extends along at least a portion of the upper and lower implant surfaces. The side walls extending between the front and rear walls generally have a minimal solid structure, for example, a plurality of struts extending between the upper and lower rims, but otherwise have open area therebetween in which the integral porous structure is formed. The configuration of the solid structure is selected to provide the implant sufficient structural integrity and mechanical stability while maximizing the area of porous structure which facilitates better integration/incorporation with the adjacent bone. In several embodiments of the disclosure, the solid structure generally encases the corners of the porous structure or otherwise houses the porous structure therein to maintain the structural integrity of the porous structure.

[0134] Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all ranges broadly recited in this document include within their scope all narrower ranges which fall within the broader ranges. It is also intended that the components of the various devices disclosed above may be combined or modified in any suitable configuration.

Claims

1. A lateral lumbar intervertebral implant for implantation in an intervertebral space between vertebrae, the implant comprising: a body having a tapered front end, a rear end and opposed side walls extending between the ends, the body having an outer perimeter with a rectangular configuration and an inner perimeter about an internal chamber, the body including: an upper surface and a lower surface, the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim; a solid front wall extending at the front end between at least the solid upper outer rim and the solid lower outer rim; a solid rear wall extending at the rear end between the solid upper outer rim and the solid lower outer rim; each of the side walls including at least one solid support strut extending between the upper outer rim and the lower outer rim; a plurality of upper solid cross struts connected between the upper outer rim and the upper inner rim and arranged in parallel to each other; a plurality of lower solid cross struts connected between the lower outer rim and the lower inner rim and arranged in parallel to each other and a porous structure integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and the solid struts in the side walls, the porous structure extending from the body outer perimeter to the body inner perimeter.

2. The intervertebral implant of claim 1, wherein the body has a wedge configuration tapering from a thicker height along the outer edge of one of the side walls to a thinner height along the outer edge of the other one of the side walls.

3. The intervertebral implant of claim 1, wherein the at least one solid support strut of each of the sidewalls includes a single support strut extending perpendicularly between the upper outer rim and the lower outer rim at an approximate mid-point of the side wall.
4. The intervertebral implant of claim 1, wherein the at least one solid support strut extends perpendicularly relative to the upper outer rim and the lower outer rim.
5. The intervertebral implant of claim 1, wherein the plurality of upper solid cross struts are connected perpendicularly between the upper outer rim and the upper inner rim.
6. The intervertebral implant of claim 1, wherein each of the cross struts defines a portion of a surface serration.
7. The intervertebral implant of claim 6, wherein all of the solid support struts and cross struts are embedded within the porous structure.
8. The intervertebral implant of claim 1, wherein the tapered front end includes a solid structure having an upper sloped portion and a lower sloped portion together defining the taper.
9. The intervertebral implant of claim 1, wherein the rear end is defined substantially by a solid structure.
10. The intervertebral implant of claim 9, wherein the solid structure defined by the rear end includes a recess for receiving an insertion instrument.
11. The intervertebral implant of claim 1, wherein the body further includes a solid secondary rear wall extending between the upper inner rim and the lower inner rim.
12. The intervertebral implant of claim 1, wherein the upper inner rim and the lower inner rim are both a closed rim.
13. The intervertebral implant of claim 1, wherein the porous structure has a greater than 60% porosity.
14. The intervertebral implant of claim 1, wherein the porous structure has an average pore size of about 300-800 μm .
15. The intervertebral implant of claim 1, wherein the porous structure has a greater than 60% porosity and has an average pore size of about 300-800 μm .
16. A lateral lumbar intervertebral implant for implantation in an intervertebral space between vertebrae, the implant comprising: a body having a tapered front end, a rear end and opposed side walls extending between the ends, the body having an outer perimeter with a rectangular configuration and an inner perimeter about an internal chamber, the body including: an upper surface and a lower surface, the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim; a solid front wall extending at the front end between at least the solid upper outer rim and the solid lower outer rim; a solid rear wall extending at the rear end between the solid upper outer rim and the solid lower outer rim; each of the side walls including at least one solid support strut extending between the upper outer rim and the lower outer rim; a plurality of upper solid cross struts connected between the upper outer rim and the upper inner rim and arranged in parallel to each other; a plurality of lower solid cross struts connected between the lower outer rim and the lower inner rim and arranged in parallel to each other and a porous structure integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and the solid struts in the side walls, the porous structure extending from the body outer perimeter to the body inner perimeter, the porous structure having a greater than 60% porosity and has an average pore size of about 300-800 μm .
17. The intervertebral implant of claim 16, wherein the at least one solid support structure of each of the sidewalls includes a single support strut extending perpendicularly between the upper outer rim and the lower outer rim at an approximate mid-point of the side wall.
18. The intervertebral implant of claim 16, wherein the body further includes a solid secondary rear wall coupled between the upper inner rim and the lower inner rim.
19. The intervertebral implant of claim 16, wherein the upper inner rim and the lower inner rim are

both a closed rim.

20. The intervertebral implant of claim 16, wherein the at least one solid support strut extends perpendicularly relative to the upper outer rim and the lower outer rim.
